FRONTEND SIMPLIFIED

Git & GitHub Cheat Sheet

Every command from the Git lessons, in the order you'll use them. Your safety net, your time machine, and a record of every change.

1 · Why Git + Setup 2 · Init, Stage & Commit 3 · Push, Pull & Deploy

01 Why version control?

Without Git: the folder-name graveyard

project-final project-FINAL-v2

project-FINAL-FOR-REAL

Lost work, emailed zip files, teammates overwriting each other, and no way to answer "what changed since Tuesday?"

With Git: a structured history

a1b2c3d Add contact form Jon · Thu 9f8e7d6 Fix broken footer link Jon · Tue

Every change gets a name, author, timestamp and message. About 96% of developers use Git.

02 Git vs GitHub: not the same thing

	Git	GitHub
What it is	A tool that lives on your computer	A website that hosts your code online
What it does	Tracks changes and history, offline	Backs up, shares, collaborates
Think of it as	The engine. Like the email protocol	The garage. Like Gmail

You can use Git without GitHub, but not GitHub without Git. Other Git hosts: GitLab, Bitbucket, Azure DevOps.

03 One-time setup (do this once, ever)

1 Create a GitHub account

github.com → Sign Up → verify your email with the code.

Heads up: your username is public and in every project URL. Keep it professional.

2 Install Git

Windows: git-scm.com installer. Keep defaults, but set the editor to **VS Code** and the branch name to **main**.

Mac: install Homebrew from **brew.sh** first, then run `brew install git`

3 Tell Git who you are

```
git config --global user.name "Your Name"
git config --global user.email "you@mail.com"
git config --global init.defaultBranch main
```

Use your **GitHub email**. It's a name tag, not a login.
`--global` = every project.

4 Connect VS Code to GitHub

Accounts icon (bottom left) → Sign in to GitHub → authorize in the browser. No more logging in on every push.

Check: `git --version` shows a number.
`git config --global user.name` shows your name.

"git" not recognized? Open a fresh terminal.

BONUS Not covered in the lessons yet, but good to know. 1 / 3

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Git & GitHub Cheat Sheet INIT, STAGE & COMM

Desktop/
fes-projects/ container only
my-first-repo/ open THIS
index.html
style.css

Start fresh: no .git folder anywhere above your project, or Git latches onto the wrong one.

Don't	Do (kebab-case)
`My First Repo`	`my-first-repo`
`MyFirstRepo`	`my-first-repo`

Lowercase, dashes instead of spaces. Wrong casing breaks React and causes 404s on GitHub Pages.

Open the right folder: VS Code → Open Folder → select **my-first-repo**. Hover its name in the VS Code sidebar: new file / new folder icons mean it's the root.

Open the terminal: View → New Terminal, or right-click `index.html` → Open in Integrated Terminal. The path should end in `/my-first-repo`.

05 git init: flip the switch

Don't Do (kebab-case) My First Repo

"not a git repository"

Expected! Git isn't watc

git init

git status

"Initialized empty

ty Git repository"
ning this folder

→

`git status`
"On branch main"
New files show as **untracked**

`git status` **is your go-to.** Not sure what's happening? Run it. The Source Control badge in VS Code shows the same info.

Never delete the hidden .git folder. It holds your entire history. VS Code hides it, and that's normal.

06 The three zones: how Git saves

Your files as you edit them in VS Code

The whole school hanging out in the gym

git add .

ZONE 1
Working Directory

ZONE 2
Staging Are

Files picked f

rea
for your next
ed up for the

`git commit` →

ZONE 3
Repository
Your permanent saved history
The photo, in the yearbook forever

The loop (every time you save)

git status changed? git add . stage all git status staged? git commit -m "Add navbar" save
git status clean?

Goal: nothing to commit, working tree clean . One file only? git add index.html

Good commit messages

Don't	Do
`"update"`	"Add navigation bar to homepage"
`"stuff"`	"Fix broken link in footer"
`"asdf"`	"Add index.html with starter content"

Watching is not saving. Untracked files aren't protected, and Live Server updating means nothing to Git. Finish the loop.

Good commit messages

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Git & GitHub Cheat Sheet PUSH, PULL & DEPLOY 07 Connect to GitHub and push (once per project)

On github.com

1 Click + New repository

2 Name it the same as your folder

3 Set it to Public. Leave the rest alone 4 Create, then copy the repo URL

Keep it empty: checking "Add a README" creates a commit that makes your first push fail.

and git pull just work. Refresh GitHub to see your files.

08 Every update after that

git add .
git commit -m "feat: add contact form" git push
In your VS Code terminal

git init if you haven't yet git remote add origin <your-url> link to GitHub git remote -v check the link git add .
git commit -m "initial commit"
git push -u origin main first push only

-u links your main to GitHub's, so from now on plain git push

09 Read your history

git log Full history with authors git log --oneline One line per commit

git show a1b2c3d One commit's changes
git push sends your commits up. git pull brings GitHub's commits down (it's fetch + merge in one).

git diff BONUS Unstaged changes Stuck inside a long log? Press q to quit. BONUS

10 Conventional commits: type(scope): description

feat New feature	fix Bug fix	Docs only docs	Restructure, same behavior refactor	Add or update tests test	Tools, dependencies chore

About 50 characters, imperative ("add" not "added"), no period. Atomic: one idea per commit, not one file or one hour.

git commit -m "fix(cart): prevent duplicate items" git commit -m "Subject" -m "Longer body" 2 lines

11 Go live with GitHub Pages

Repo **Settings** → **Pages** (left sidebar) → Branch: **None → main** → **Save** → Wait, refresh: **live URL**

Share code: green **Code** button. **Share site:** the Pages URL. Homepage must be `index.html`.

12 When things go wrong

You see	What it means → fix
`not a git repository`	Wrong folder, or no `git init` yet. Open the project folder itself
`Updates were rejected`	GitHub has commits you don't. Run `git pull`, then `git push`
First push rejected right away	The repo was made with a README. Pull first, or recreate it empty
Password rejected	No passwords allowed. Sign in via VS Code or `gh auth login`
`git` not recognized	Close the terminal and open a fresh one

When in doubt: **git status**. It always tells you what Git knows.

Repo Settings Pages (left sidebar) Branch: None main Save Wait, refresh: live URL Share code: green Code button. Share site: the Pages URL. Homepage must be index.html .

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